

# PAW on-site valence exact exchange: CoQuí vs. the correct (all-electron) expression

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## Abstract

This note documents (i) exactly what CoQuí currently computes for the PAW on-site valence exchange, term by term, from the source; (ii) what ABINIT computes and why it equals the true all-electron (AE) exchange; and (iii) a derivation of the difference between the two, which turns out to be a single, physically identifiable term that CoQuí neglects. The analysis is backed by the instrumented Si  $a = 10.20$  comparison (both codes, finite-size correction off, on-site term toggled): CoQuí on-site exchange +0.0414 Ha vs. ABINIT +0.0925 Ha (converged), a 0.052 Ha gap that the derived missing term accounts for.

## Source of record.

- CoQuí smooth + augmentation + one-center: `src/hamiltonian/paw/v_x_paw.hpp`
- CoQuí deltaC kernel: `src/python/mean_field/abinit_interface/paw_deltaC.py` (identical definition to QE's `pw2coqui`; stored in the mean-field h5 and read into `species_paw_t::deltaC`)
- ABINIT on-site oscillator: `abinit-10.6.7/src/65_paw/m_pawpwij.F90`, used by `calc_sigx_me (src/70_gw/m_sigx.F90)`

## 1 Notation

Per  $k$ -point pair  $(m, k_q)$ ,  $(n, k_p)$  the *pair (transition) density* is  $\rho_{mn}(\mathbf{r}) = \psi_{mk_q}^*(\mathbf{r}) \psi_{nk_p}(\mathbf{r})$ . The exact-exchange energy is the bilinear

$$E_x = -\frac{1}{2} \sum_s \sum_{k_q k_p} \sum_{mn} \frac{f_{mk_q} f_{nk_p}}{N_k} \langle \rho_{mn} | v_C | \rho_{mn} \rangle, \quad \langle \rho | v_C | \rho' \rangle \equiv \frac{4\pi}{\Omega} \sum_G \frac{\rho^*(K) \rho'(K)}{|K|^2}, \quad K = G + k_p - k_q. \quad (1)$$

The matrix-element form actually built in the code is  $K_{ij}(s, k_p) = -\frac{1}{N_k} \sum_{k_q} \sum_n f_{nk_q} \langle \rho_{ni} | v_C | \rho_{nj} \rangle$ , with  $E_x = \frac{1}{2} \sum_i f_i K_{ii}$ .

**PAW decomposition of the pair density.** With smooth orbitals  $\tilde{\psi}$ , projectors  $p_I^a$ , and AE/PS partial waves  $\phi_I^a, \tilde{\phi}_I^a$ :

$$\rho_{mn}(\mathbf{r}) = \underbrace{\tilde{\rho}_{mn}(\mathbf{r})}_{\text{smooth (grid)}} + \sum_a \underbrace{[\rho_{mn,a}^1(\mathbf{r}) - \tilde{\rho}_{mn,a}^1(\mathbf{r})]}_{D_{mn,a}(\mathbf{r}) \text{ (on-site, radial)}}, \quad (2)$$

$$\rho_{mn,a}^1 = \sum_{IJ} P_{mI}^{a*} P_{nJ}^a \phi_I^a \phi_J^a, \quad \tilde{\rho}_{mn,a}^1 = \sum_{IJ} P_{mI}^{a*} P_{nJ}^a \tilde{\phi}_I^a \tilde{\phi}_J^a, \quad P_{nJ}^a \equiv \langle p_J^a | \tilde{\psi}_n \rangle. \quad (3)$$

The *compensation charge*  $\hat{n}_{mn,a}$  reproduces the *multipole moments* of  $D_{mn,a}$  but not its shape:

$$\hat{n}_{mn,a}(\mathbf{r}) = \sum_{IJ} P_{mI}^{a*} P_{nJ}^a \sum_L q_{IJ}^L g_L(r) \{Y_{LM}\}, \quad q_{IJ}^L = \int (\phi_I \phi_J - \tilde{\phi}_I \tilde{\phi}_J) r^L r^2 dr, \quad (4)$$

with  $g_L$  the (Bessel/Gaussian) shape function normalised to unit  $L$ -moment. By construction  $\int (D_{mn,a} - \hat{n}_{mn,a}) r^L Y_{LM} d^3r = 0$  for all  $LM$ .

Two identities used below (Coulomb bilinearity + the moment property):

$$\langle D | v_C | D \rangle - \langle \rho^1 | v_C | \rho^1 \rangle + \langle \tilde{\rho}^1 | v_C | \tilde{\rho}^1 \rangle = -2 \langle \tilde{\rho}^1 | v_C | D \rangle \quad (\text{expand } D = \rho^1 - \tilde{\rho}^1). \quad (5)$$

## 2 What CoQuí currently implements

CoQuí builds  $K_{ij}$  as **three** additive pieces (`v_x_paw.hpp`, function `v_x`).

**(1a) Smooth pair density (grid).** Lines 442–446. For occupied  $n$  at  $k_q$  and every  $a$  at  $k_p$ :

$$\tilde{\rho}_{na}(G) = \text{FFT}[\tilde{u}_n^*(k_q, \mathbf{r}) \tilde{u}_a(k_p, \mathbf{r})](G). \quad (6)$$

**(1b) Compensation-charge augmentation (grid), lines 448–485.**

$$\rho_{na}^{\text{full}}(G) = \tilde{\rho}_{na}(G) + \Omega \sum_a \sum_{IJ} P_{nI}^{a*}(k_q) P_{aJ}^a(k_p) Q_a^{IJ}(K) e^{-iK \cdot \tau_a} \quad (7)$$

where  $Q_a^{IJ}(K)$  is the Fourier transform of the **compensation charge**  $\hat{n}_{IJ}$  (moments  $\times$  shape function), carrying the  $4\pi/\Omega$  prefactor (`evaluate_Q_IJ_at_K_fast`, l.229; the explicit  $\Omega$  on l.461 restores the density normalisation). Crucially this is  $\hat{n}$ , i.e. **moments only** — it is *not* the AE–PS pair density. So on the grid CoQuí forms  $\rho^{\text{full}} = \tilde{\rho} + \hat{n}$ .

**(1c) Coulomb contraction (smooth + augmentation), lines 487–497.**

$$K_{ij}^{\text{sm+aug}}(s, k_p) = -\frac{1}{N_k \Omega} \sum_{k_q} \sum_n f_{nk_q} \sum_G v_C(K) \rho_{ni}^{\text{full}*}(G) \rho_{nj}^{\text{full}}(G), \quad \left[ \text{scl} = -\frac{1}{N_k \Omega} \right]. \quad (8)$$

In bilinear form this is  $-\frac{1}{N_k} \sum_{k_q} \sum_n f_n \langle \tilde{\rho} + \hat{n} | v_C | \tilde{\rho} + \hat{n} \rangle$ .

**(1d) One-center deltaC correction, lines 503–590.**

$$\Delta K_{ij}^{\text{oc}}(s, k_p) = -\frac{1}{N_k} \sum_{k_q} \sum_n f_{nk_q} \sum_a \sum_{IJKL} P_{iI}^{a*}(k_p) P_{nJ}^a(k_q) \Delta C^a(I, J, K, L) P_{nK}^{a*}(k_q) P_{jL}^a(k_p), \quad (9)$$

with  $\text{scl}_{\text{oc}} = -\frac{1}{N_k}$  and (`paw_deltaC.py`, l.111–142; QE/pw2coqui identical):

$$\Delta C^a(I, J, K, L) = K^{\text{AE}} - K^{\text{PS}}, \quad K = \sum_{LM} \int V_{LM}^{IJ}(r) \rho_{LM}^{KL}(r) dr, \quad (10)$$

$$\rho_{LM}^{IJ,AE} = \sum_{lm} \text{ap}(lm, l_i, l_j) \underbrace{\phi_I \phi_J}_{\text{pfunc}}, \quad (11)$$

$$\rho_{LM}^{IJ,PS} = \sum_{lm} \text{ap}(lm, l_i, l_j) \left[ \underbrace{\tilde{\phi}_I \tilde{\phi}_J}_{\text{ptfunc}} + \underbrace{q_{IJ}^L g_L(r)}_{\text{qfunc}=\hat{n}} \right]. \quad (12)$$

**Read literally:** the AE side is the pure product  $\phi_I \phi_J$ ; the PS side is  $\tilde{\phi}_I \tilde{\phi}_J + \hat{n}_{IJ}$  (note the + `qfunc1` on l.107/126 of `paw_deltaC.py` and l.215 of `paw_onecenter.hpp`). Hence, per atom,

$$\Delta C = \langle \phi \phi | v_C | \phi \phi \rangle^{\text{AE}} - \left\langle \tilde{\phi} \tilde{\phi} + \hat{n} \middle| v_C \middle| \tilde{\phi} \tilde{\phi} + \hat{n} \right\rangle^{\text{PS}} = \langle \rho^1 | v_C | \rho^1 \rangle - \langle \tilde{\rho}^1 + \hat{n} | v_C | \tilde{\rho}^1 + \hat{n} \rangle. \quad (13)$$

**(1e) CoQuí total (compensated PAW).** Collecting (1c)+(1d) and cancelling the shared  $\langle \hat{n} | v_C | \hat{n} \rangle$ :

$$E_x^{\text{CoQuí}} = -\frac{1}{2} \sum \frac{ff}{N_k} \left[ \langle \tilde{\rho} | v_C | \tilde{\rho} \rangle + 2 \langle \tilde{\rho} | v_C | \hat{n} \rangle - 2 \langle \tilde{\rho}^1 | v_C | \hat{n} \rangle + \langle \rho^1 | v_C | \rho^1 \rangle - \langle \tilde{\rho}^1 | v_C | \tilde{\rho}^1 \rangle \right] \quad (14)$$

This is the standard **compensated (moment-matched) PAW** exchange: the smooth density interacts with the *multipoles*  $\hat{n}$  of the on-site AE–PS difference, plus an exact on-site AE–(PS+comp) Fock correction.

*Prefactor note:* `scl` =  $-1/(N_k \Omega)$  for the grid Coulomb measure and `scloc` =  $-1/N_k$  for the already-integrated radial `deltaC` are internally consistent and validated to machine precision against CoQuí’s own THC  $K_a$  (`test vx_onecenter_vs_thc_Ka`). The prefactors are **not** the issue here.

### 3 What ABINIT computes (= the correct AE exchange)

ABINIT builds the **oscillator** of the *full AE pair density* directly (`m_pawpwij.F90`). The on-site addition to  $\langle \tilde{\psi}_m | e^{-iK\mathbf{r}} | \tilde{\psi}_n \rangle$  is (l.479, 527, 561, 859–918):

$$M_{mn}^{\text{onsite}}(K) = \sum_a \sum_{IJ} P_{mI}^{a*} P_{nJ}^a q_{IJ}^a(K), \quad q_{IJ}^a(K) = 4\pi \sum_{LM} (-i)^L Y_{LM}(\hat{K}) G_{l_i m_i, l_j m_j}^{LM} \text{ff}_{IJ}^{aL}(|K|) \quad (15)$$

$$\text{ff}_{IJ}^{aL}(K) = \int_0^{r_a} j_L(Kr) [\phi_I \phi_J - \tilde{\phi}_I \tilde{\phi}_J](r) dr \quad (16)$$

i.e. the transform of the **full AE–PS on-site pair density  $D$ , with the true radial shape and NO compensation charge** (`phiphj - tphitphj`, l.561; the `use_pawnhat0 = 0` default in `m_sigx.F90`:l.155). The exchange is then

$$E_x^{\text{ABINIT}} = -\frac{1}{2} \sum \frac{ff}{N_k} \langle \tilde{\rho} + D | v_C | \tilde{\rho} + D \rangle = -\frac{1}{2} \sum \frac{ff}{N_k} \left[ \langle \tilde{\rho} | v_C | \tilde{\rho} \rangle + 2 \langle \tilde{\rho} | v_C | D \rangle + \langle D | v_C | D \rangle \right]. \quad (17)$$

Because  $\tilde{\rho} + D$  is the exact AE pair density (up to the `ecutsigx` truncation of the sharp  $D$ , which we verified converged: on-site moves  $< 1$  mHa from `ecutsigx` = 25  $\rightarrow$  50), this equals the true AE exchange. This is the Arnaud–Alouani / Shishkin–Kresse PAW–GW oscillator.

## 4 The difference (derivation)

Subtract § (1e) from §3, per pair, using identity (5):

$$E_x^{\text{ABINIT}} - E_x^{\text{CoQuí}} = -\frac{1}{2} \sum \frac{f f}{N_k} \left[ 2 \langle \tilde{\rho} | v_C | D \rangle - 2 \langle \tilde{\rho} | v_C | \hat{n} \rangle + 2 \langle \tilde{\rho}^1 | v_C | \hat{n} \rangle - 2 \langle \tilde{\rho}^1 | v_C | D \rangle \right] \quad (18)$$

$$E_x^{\text{ABINIT}} - E_x^{\text{CoQuí}} = - \sum \frac{f_m f_n}{N_k} \langle \tilde{\rho}_{mn} - \tilde{\rho}_{mn}^1 | v_C | D_{mn} - \hat{n}_{mn} \rangle \quad (19)$$

This is the term CoQuí drops. Its two factors are each localised in/near the augmentation sphere:

- $\tilde{\rho} - \tilde{\rho}^1$  — the smooth pair density minus its on-site PS partial-wave expansion. Vanishes only if the PS partial waves are *complete* inside the sphere (the usual PAW assumption  $\tilde{\rho}|_{\text{sphere}} = \tilde{\rho}^1$ ).
- $D - \hat{n}$  — the AE–PS on-site difference minus its multipole moments: the **shape of  $D$  beyond its moments**. Zero moments by construction, nonzero shape.

The compensated PAW total-energy formalism sets this cross term to zero. That is an excellent approximation for the *Hartree/total energy* (long-ranged, moment dominated) but **not** for **exchange**, which is short-ranged and sensitive to the pair-density shape inside the core. For Si the numbers are:

| quantity (Si $a = 10.20$ , $4 \times 4 \times 4$ , finite-size off)                                  | value (Ha)               |
|------------------------------------------------------------------------------------------------------|--------------------------|
| smooth only, both codes (identical to 6 $\mu\text{Ha}$ )                                             | −1.77964                 |
| ABINIT on-site contribution $2 \langle \tilde{\rho}   v_C   D \rangle + \langle D   v_C   D \rangle$ | <b>+0.0925</b>           |
| CoQuí $\langle \tilde{\rho}   v_C   \hat{n} \rangle$ -type augmentation (1b/1c extra)                | +0.0481                  |
| CoQuí <b>deltaC</b> one-center (1d)                                                                  | −0.0067                  |
| CoQuí on-site total                                                                                  | <b>+0.0414</b>           |
| <b>missing cross term</b> $-\langle \tilde{\rho} - \tilde{\rho}^1   v_C   D - \hat{n} \rangle$       | $\approx$ <b>+0.0511</b> |

The sign is consistent: CoQuí’s on-site is too small (recovers  $\sim 44\%$ ), so its total exchange is too *negative* by 0.052 Ha (the missing term is repulsive/less-binding).

## 5 What I believe is the correct expression

Two mathematically equivalent routes recover the missing term. Both are “correct”; they differ in numerical convenience.

### Route A — shape-restored oscillator (ABINIT/Arnaud–Alouani; recommended)

Replace CoQuí’s compensation-charge augmentation (§ 1b) with the **full AE–PS pair-density form factor**, and then **drop deltaC entirely**:

$$\rho_{na}^{\text{full}}(G) = \tilde{\rho}_{na}(G) + \sum_a \sum_{IJ} P_{nI}^{a*}(k_q) P_{aJ}^a(k_p) \mathcal{Q}_a^{IJ}(K) e^{-iK\tau_a}, \quad \mathcal{Q}_a^{IJ}(K) = \int_0^{r_a} j_L(Kr) [\phi_I \phi_J - \tilde{\phi}_I \tilde{\phi}_J] dr \times (\text{ang.}), \quad (20)$$

i.e. use `pfunc - ptfunc` (AE–PS product) in the radial Hankel transform *instead of* `qfunc1` (the compensation moments). Structurally this is a one-line change to `build_grad_tab / evaluate_Q_IJ_at_K`: feed `pfunc - ptfunc` rather than the `moment×shape` `qfunc1`. Then  $K^{\text{sm+aug}}$  alone equals the exact AE exchange and the `deltaC` block (§ 1d) must be removed (it would double count).

- **Pro:** exact, matches ABINIT/VASP, no separate one-center kernel.
- **Con:**  $D$  is sharp; the augmentation FFT mesh (`fft_mesh_aug`) must resolve  $\text{ff}_{IJ}^L(K)$  out to the same effective cutoff ABINIT uses (`ecutsigx`; here converged by  $\sim 50$  Ha). This is the reason the compensated route was chosen originally —  $\hat{n}$  is smooth and grid-friendly.

### Route B — keep compensation, add the missing cross term to `deltaC`

Keep § 1b/1c as is, but redefine the one-center kernel so that it also supplies  $-\langle \tilde{\rho} - \tilde{\rho}^1 | v_C | D - \hat{n} \rangle$ . In practice this means the on-site Fock must be evaluated between the *smooth-minus-onsite* density and the *shape residual*  $D - \hat{n}$ , not only the pure on-site AE/PS products. This is harder to cast as a frozen  $n_h^4$  matrix because  $\tilde{\rho} - \tilde{\rho}^1$  depends on the actual smooth orbitals, not just the projector amplitudes — so Route A is cleaner.

### Recommendation

Implement **Route A**: swap the augmentation form factor from the compensation charge (`qfunc1`, moments) to the full AE–PS partial-wave product (`pfunc - ptfunc`, shape), on a mesh converged like ABINIT’s `ecutsigx`, and drop `deltaC` from the exchange. Validate against the numbers here (Si on-site must go  $+0.0414 \rightarrow +0.0925$  Ha; total vv exchange  $-1.7382 \rightarrow -1.6863$  Ha; per-band  $\Sigma_x$  must match ABINIT `iszoff` band-for-band). Keep the compensation-charge path for the **Hartree** `deeq` (where it is correct and grid-friendly).

**Caveat / next check.** The identity in §4 assumes the exchange is the clean bilinear  $\langle \rho_{mn} | v_C | \rho_{mn} \rangle$  and that CoQu’s terms map exactly as written above; the 0.052 Ha numerical closure supports this. Before implementing, reproduce the decomposition at a second system (e.g. LiH or a 3d oxide, where on-site exchange is larger) to confirm the missing-term identity holds beyond Si *sp*.